

## Unit 112 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 80.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 40 plots, 428 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                    | STAND INFO          |                                                |
|----------------------|------------------------------------|---------------------|------------------------------------------------|
| Basal area           | 214 sq ft/ac (SE 2.5 %)            | Type                | Sierran mixed conifer                          |
| Trees per acre       | 92                                 | CWHR size / density | 4 / D (cover 87 %)                             |
| QMD                  | 20.6 in                            | SDI                 | 279 (48 % of max 580)                          |
| SAWTIMBER (>= 10 in) |                                    | BIOMASS (< 10 in)   |                                                |
| BA   QMD             | 210 sq ft/ac   21.5 in             | BA   QMD            | 4 sq ft/ac   8.6 in                            |
| TPA                  | 84                                 | TPA                 | 9                                              |
| Volume               | 44.4 MBF/ac (8077 cu ft)           | Volume              | 2.2 green tons/ac (72 cu ft)                   |
| TREATMENT TARGET     |                                    | CRUISE DESIGN       |                                                |
| Leave                | 35 % of max SDI = ~156 sq ft/ac BA | Sampling error      | 5.1 % at 95 % (t = 2.02, n = 40)               |
| Species (BA)         | WF 66, PP 63, IC 36, DF 30, SP 19  | Stratum standard    | 40 % - MEETS; 40 plots vs 20-plot practice: ok |

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 4-7.9         | 2          | 3.4         | 1.0          | 16          | 0.1         |
| 8-11.9        | 19         | 16.0        | 9.5          | 234         | 1.3         |
| 12-15.9       | 47         | 21.9        | 23.5         | 637         | 3.5         |
| 16-19.9       | 62         | 17.6        | 31.0         | 952         | 5.2         |
| 20-23.9       | 61         | 11.7        | 30.5         | 1068        | 5.9         |
| 24-27.9       | 69         | 9.5         | 34.5         | 1336        | 7.3         |
| 28-31.9       | 51         | 5.2         | 25.5         | 1066        | 5.9         |
| 32-35.9       | 42         | 3.4         | 21.0         | 926         | 5.1         |
| 36-39.9       | 28         | 1.8         | 14.0         | 680         | 3.7         |
| 40-43.9       | 12         | 0.6         | 6.0          | 296         | 1.6         |
| 44-47.9       | 12         | 0.5         | 6.0          | 307         | 1.7         |
| 48-51.9       | 9          | 0.3         | 4.5          | 247         | 1.4         |
| 52-55.9       | 9          | 0.3         | 4.5          | 248         | 1.4         |
| 56-59.9       | 1          | 0.0         | 0.5          | 26          | 0.1         |
| 60-63.9       | 4          | 0.1         | 2.0          | 111         | 0.6         |
| <b>All</b>    | <b>428</b> | <b>92.5</b> | <b>214.0</b> | <b>8149</b> | <b>44.8</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| WF         | 131        | 27.0        | 65.5         | 21.1        | 2522        | 13.9        |
| PP         | 126        | 27.2        | 63.0         | 20.6        | 2383        | 13.1        |
| IC         | 72         | 15.3        | 36.0         | 20.7        | 1415        | 7.8         |
| DF         | 61         | 14.7        | 30.5         | 19.5        | 1130        | 6.2         |
| SP         | 38         | 8.3         | 19.0         | 20.5        | 700         | 3.9         |
| <b>All</b> | <b>428</b> | <b>92.5</b> | <b>214.0</b> | <b>20.6</b> | <b>8149</b> | <b>44.8</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 112: plots on the 300 ft grid (orange = QA flag)

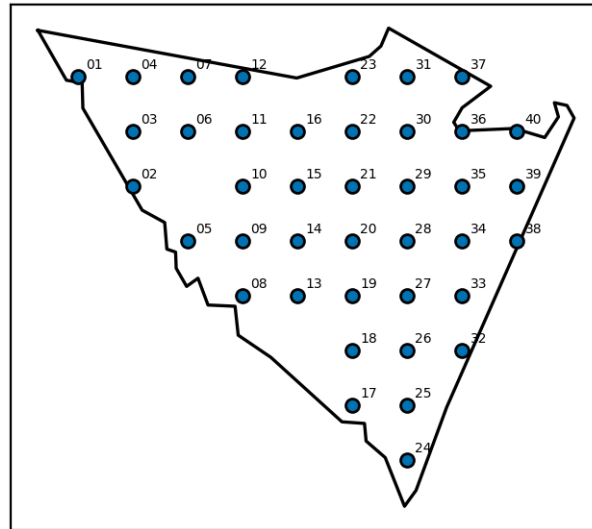


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

**Table 4. Plot list**

| <b>Plot</b> | <b>Trees in</b> | <b>BA sq ft/ac</b> | <b>Slope %</b> | <b>Status</b> |
|-------------|-----------------|--------------------|----------------|---------------|
| 112-01      | 11              | 220                | 12             | ok            |
| 112-02      | 11              | 220                | 10             | ok            |
| 112-03      | 11              | 220                | 17             | ok            |
| 112-04      | 12              | 240                | 12             | ok            |
| 112-05      | 9               | 180                | 21             | ok            |
| 112-06      | 12              | 240                | 36             | ok            |
| 112-07      | 10              | 200                | 10             | ok            |
| 112-08      | 5               | 100                | 21             | ok            |
| 112-09      | 12              | 240                | 10             | ok            |
| 112-10      | 11              | 220                | 10             | ok            |
| 112-11      | 9               | 180                | 10             | ok            |
| 112-12      | 11              | 220                | 29             | ok            |
| 112-13      | 10              | 200                | 12             | ok            |
| 112-14      | 13              | 260                | 8              | ok            |
| 112-15      | 13              | 260                | 15             | ok            |
| 112-16      | 13              | 260                | 16             | ok            |
| 112-17      | 10              | 200                | 13             | ok            |
| 112-18      | 12              | 240                | 36             | ok            |
| 112-19      | 9               | 180                | 13             | ok            |
| 112-20      | 11              | 220                | 21             | ok            |
| 112-21      | 10              | 200                | 14             | ok            |
| 112-22      | 13              | 260                | 13             | ok            |
| 112-23      | 10              | 200                | 22             | ok            |
| 112-24      | 13              | 260                | 28             | ok            |
| 112-25      | 10              | 200                | 26             | ok            |
| 112-26      | 12              | 240                | 11             | ok            |
| 112-27      | 8               | 160                | 9              | ok            |
| 112-28      | 10              | 200                | 15             | ok            |
| 112-29      | 8               | 160                | 13             | ok            |
| 112-30      | 11              | 220                | 12             | ok            |
| 112-31      | 8               | 160                | 7              | ok            |
| 112-32      | 12              | 240                | 25             | ok            |
| 112-33      | 11              | 220                | 26             | ok            |
| 112-34      | 11              | 220                | 20             | ok            |
| 112-35      | 10              | 200                | 10             | ok            |
| 112-36      | 13              | 260                | 39             | ok            |
| 112-37      | 11              | 220                | 16             | ok            |
| 112-38      | 9               | 180                | 16             | ok            |
| 112-39      | 12              | 240                | 19             | ok            |
| 112-40      | 11              | 220                | 30             | ok            |